This section discusses convergence under the stated assumptions. For each actual proximal output y, we use the true residual $r_F(y) = \inf_{v \in F(y)} \|v\|$. An error bound $d(y, S) \leq c\, r_F(y)^q$ and a reflected-resolvent estimate with exponent γ yield local contraction when $q\gamma > 1$, provided the orbit remains in the certified neighborhood. A strict length budget ensures this localization. The resulting allowed orbit has finite length and converges to a point in S.